

Q1 The following table shows the number of instructions for a program executed by two computers. (12 Marks)

	Arithmetic	Store	Load	Branch
Computer A	800	200	300	100
Computer B	500	200	650	50

The cycle per instruction (CPI) for each instruction type as the following

Instruction type	CPI
Arithmetic	4
Store	4
Load	5
Branch	3

- Assume the clock rate for computer A is 3GHZ and clock rate for computer B is 3.1 GHZ

a- Find the average cycle per instruction (CPI_{av}) when using Computer A

b- Find (Million Instructions Per Second) MIPS rating when using Computer A

c- Find the execution time when using Computer A

CNE Family

d- Fraction of total instructions executed for each instruction type when using Computer B

e- Find the average cycle per instruction (CPI_{av}) when using Computer B

f- Fraction of total execution time for each instructions type when using Computer B

g- Find the execution time when using Computer B

Q2) Consider the following C++ assignment statement:

(8 marks)

$A[i] = A[i] + x - 4;$

Assume the following:

- A is an array of integers.
 - The base address of the array A is stored in the \$t1 register.
 - The integer variable x is stored in the \$t2 register.
 - The index variable i is stored in the \$t4 register..
 - The value of OP is 43 for store instruction
 - Registers \$t0 to \$t7 have numbers from 8 to 15, respectively
- a- Write the MIPS assembly instructions necessary to implement the following C++ instruction

$A[i] = A[i] + x - 4;$

- b- For MIPS instructions you wrote, represent the last store instruction in a machine code. (the last one only)